

Team Trash

Autonomous Recycling Bin

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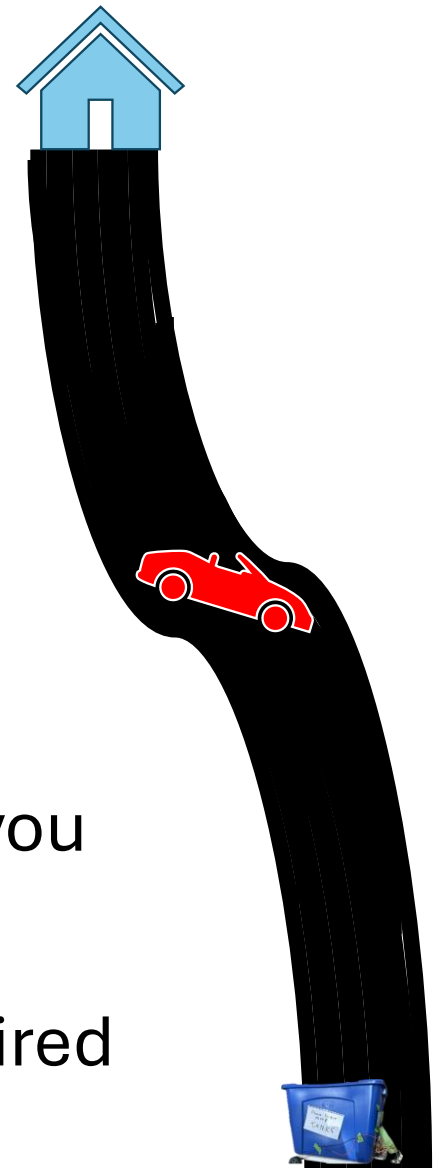


Project Overview

Problem: Who wants to walk down a long driveway to take out the recycling bin?
No one.

Solution: Build an autonomous robot to do the job for you

Beneficiaries: Lazy able-bodied, elderly/ walking impaired persons

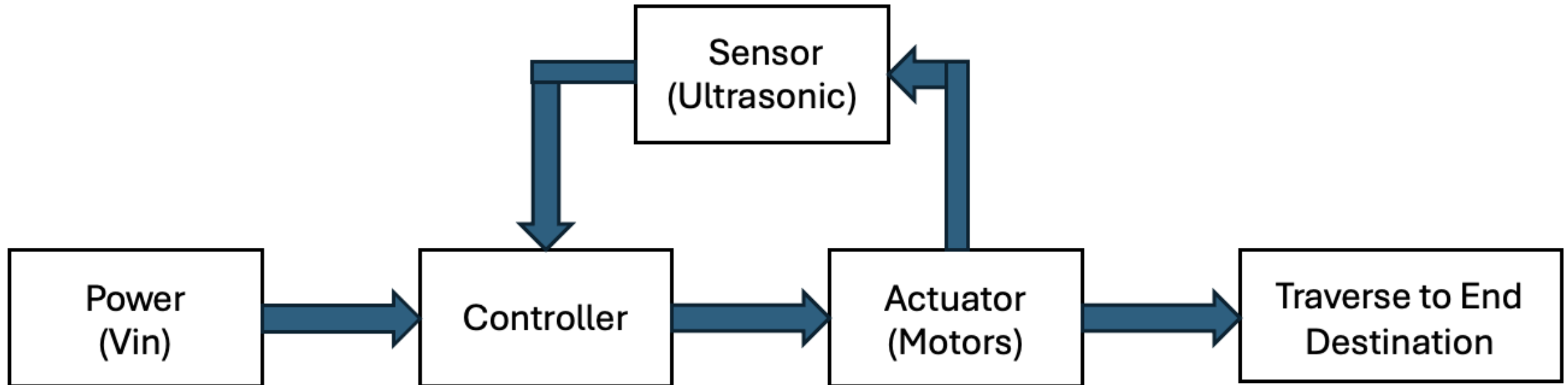


Objectives & Requirements

- Make an autonomous recycling bin that can take itself from its home location to the curb
- Maintain safety to surroundings
- Avoid collisions and maneuver around objects
- Follow a set of path to reach the end destination for pickup, navigate the path back to start location after pickup

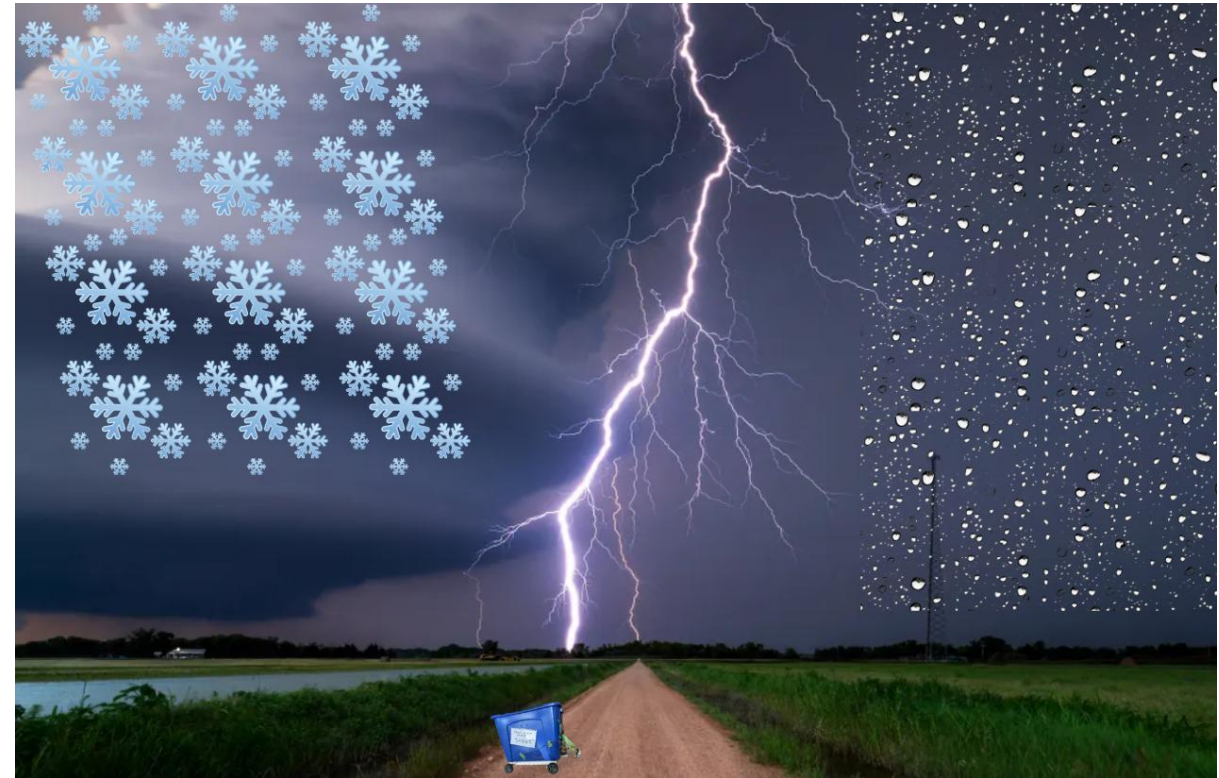


System Overview



Design Constraints / Specifications

- **Maneuverability**
 - Avoid collisions and maneuver around objects using ultrasonic sensor array
- **Weather resistant (IP 67)**
 - Heavy rain, temporary immersion, easy rinse off
 - Dust tight
- **Reach end of driveway**
 - Go from start location to end location once turned on
 - Travel walking speed (1-4) mph
- **Hold weight of recycling bin**
 - >50 lbs
- **Cost effective**
 - Working prototype assembled for under \$100



Electronics & Logic Flow

Hardware BOM

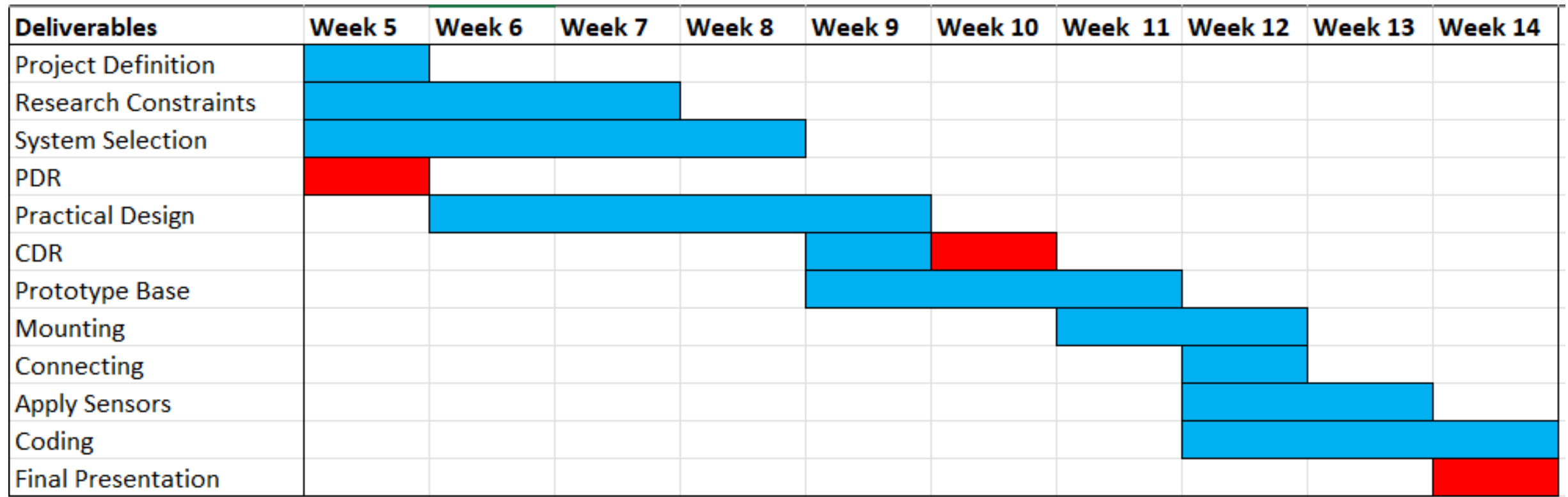
	Electronic Part	Qty and Purpose
Power	AA Batteries 1.5V	4x – supply dc power to breadboard (Teensy, sensors, motors)
Actuator	5V Servo motors	4x – torque to wheels, wired in tank drive
Controller	Arduino Teensy 4.0	1x – main controller for sensor input, logic flow, output to motors
Sensors	Ultrasonic sensors	3x – array in front of bot to detect any objects, help maneuver around and past them

Logic Flow

Control Logic (FSM)

- 1.STATE_FORWARD_NAVIGATION
- 2.STATE_OBSTACLE_DETECTED
- 3.STATE_TURN_90
- 4.STATE_SIDESTEP
- 5.STATE_REALIGN_HEADING

Gantt Chart



BOM

Part/ Item	Qty	Price	Source
Teensy 4.0 (Arduino)	1	Free	Aiden
5V Servo Stepper Motor with Wheels	4	Free	Aiden
5V Motor Wire	1 (6 feet)	\$6.36	Danbury Hobby
5V Ultrasonic sensors	3	Free	Aiden (Arduino kit)
Mounting Screws	2 (packs)	\$13.42	Hemlock Hardware Store
Recycling Bin	1	\$34.02 (tax incl)	Hemlock Hardware Store
AA Batteries	1 (8 pack)	\$12.75	Hemlock Hardware Store
Set Screws for wheels	6	\$5.55 total	Hemlock Hardware Store
Motor mounts	4	Free	Annex 3D Printers
	Total:	\$71.10	

Troubleshooting



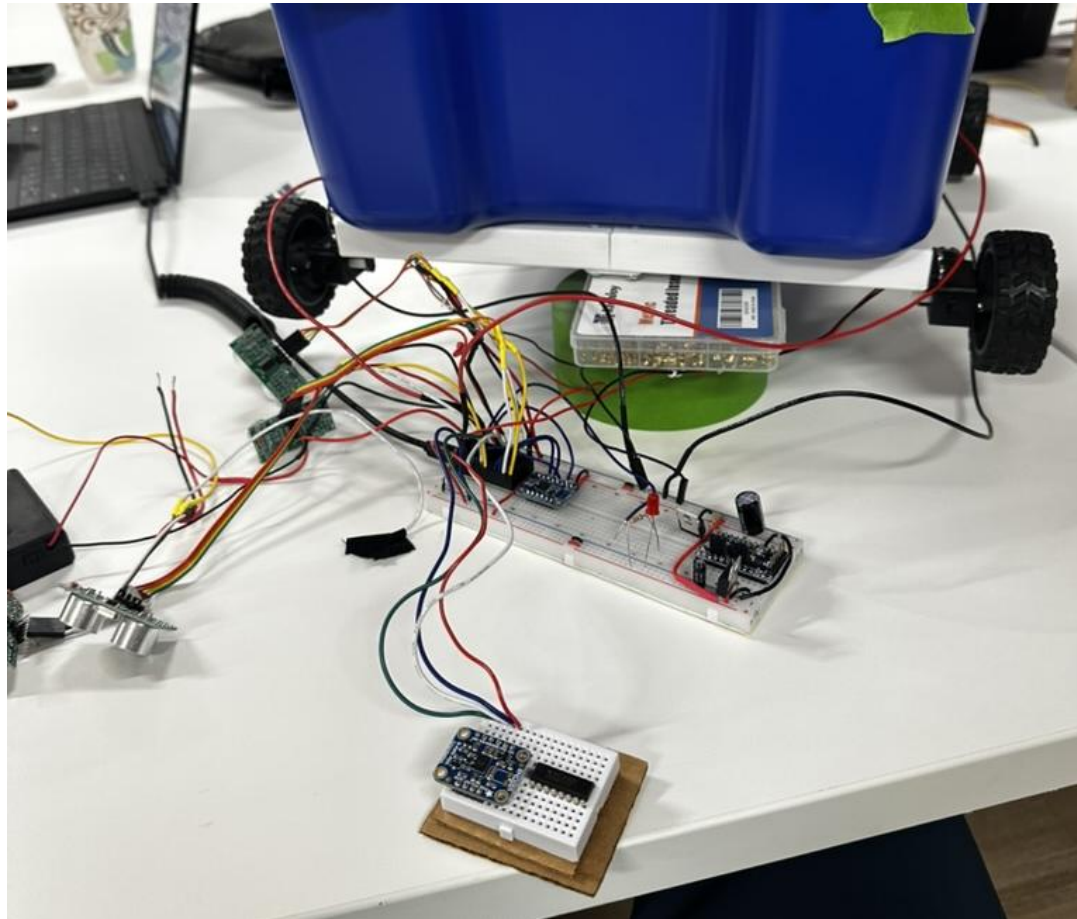
Teensy connecting and disconnecting

- Checked wiring 
- Changed computers 
- Changed batteries 

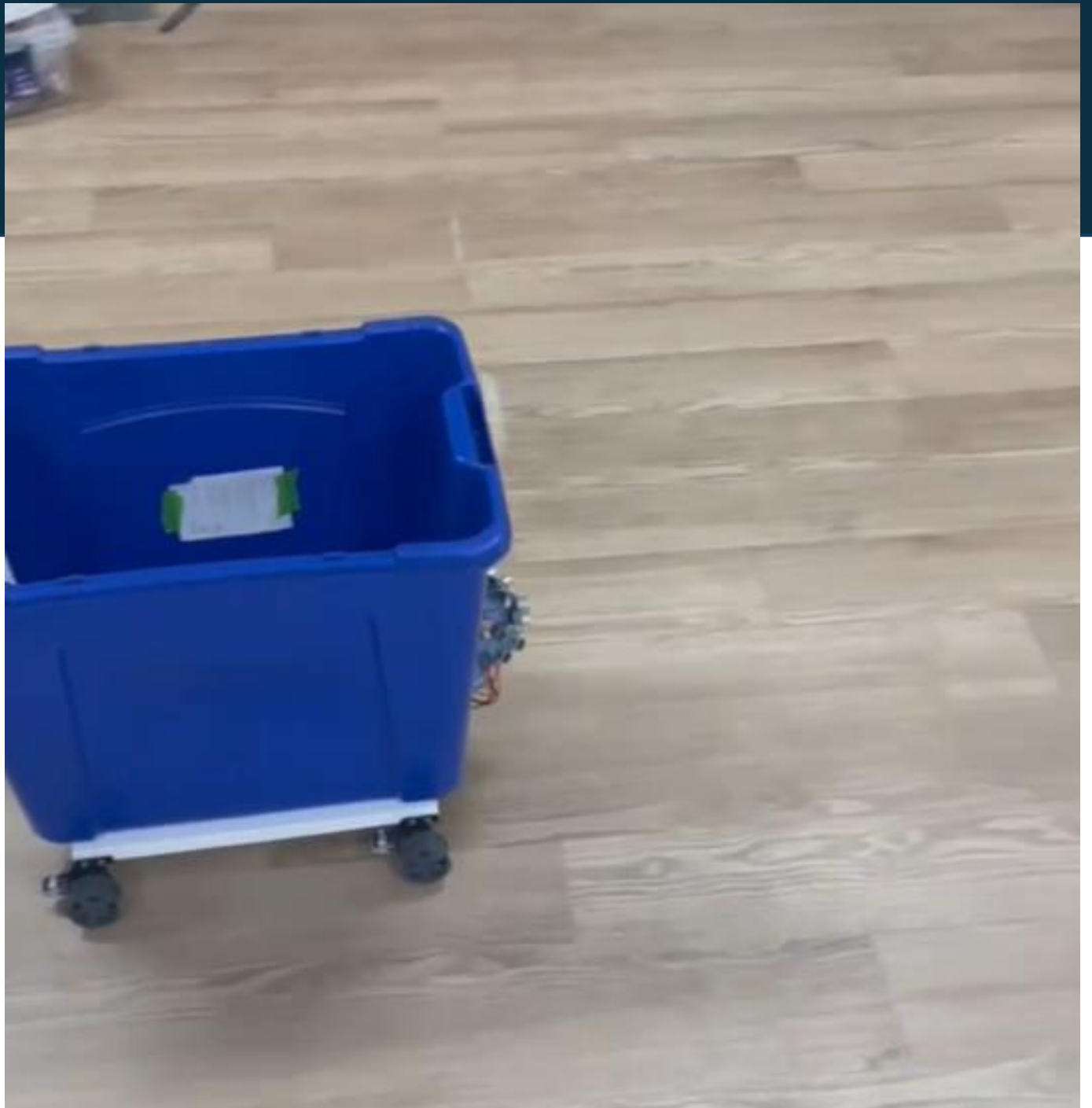
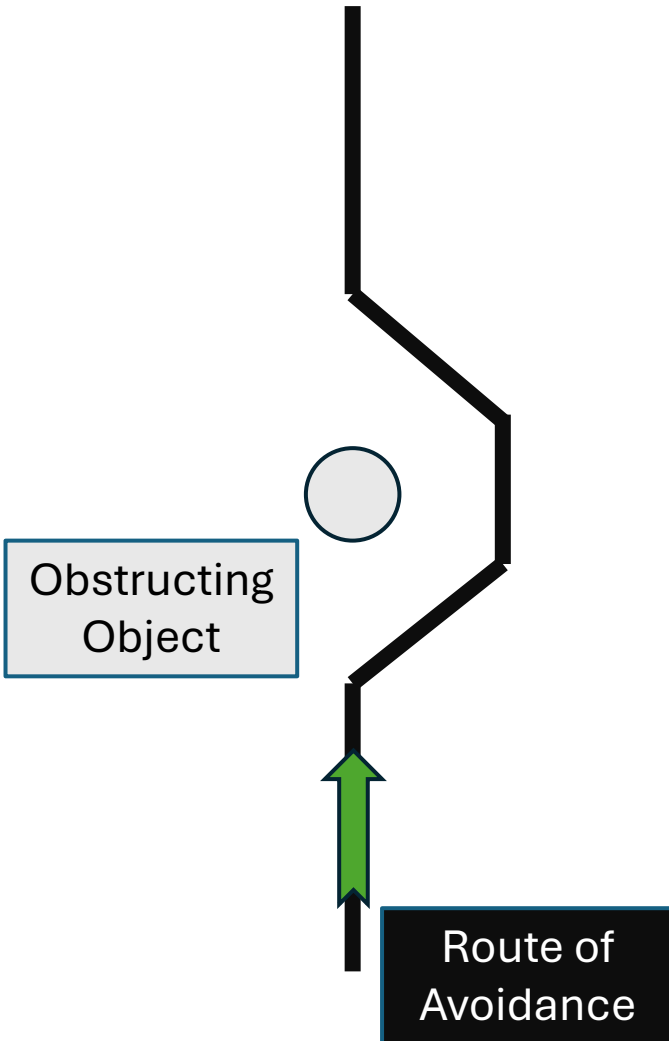
Robot unable to complete turn

- Checked and adjusted code 
- Adjusted PWM for larger weight 
- Removed rubber grip on tires 

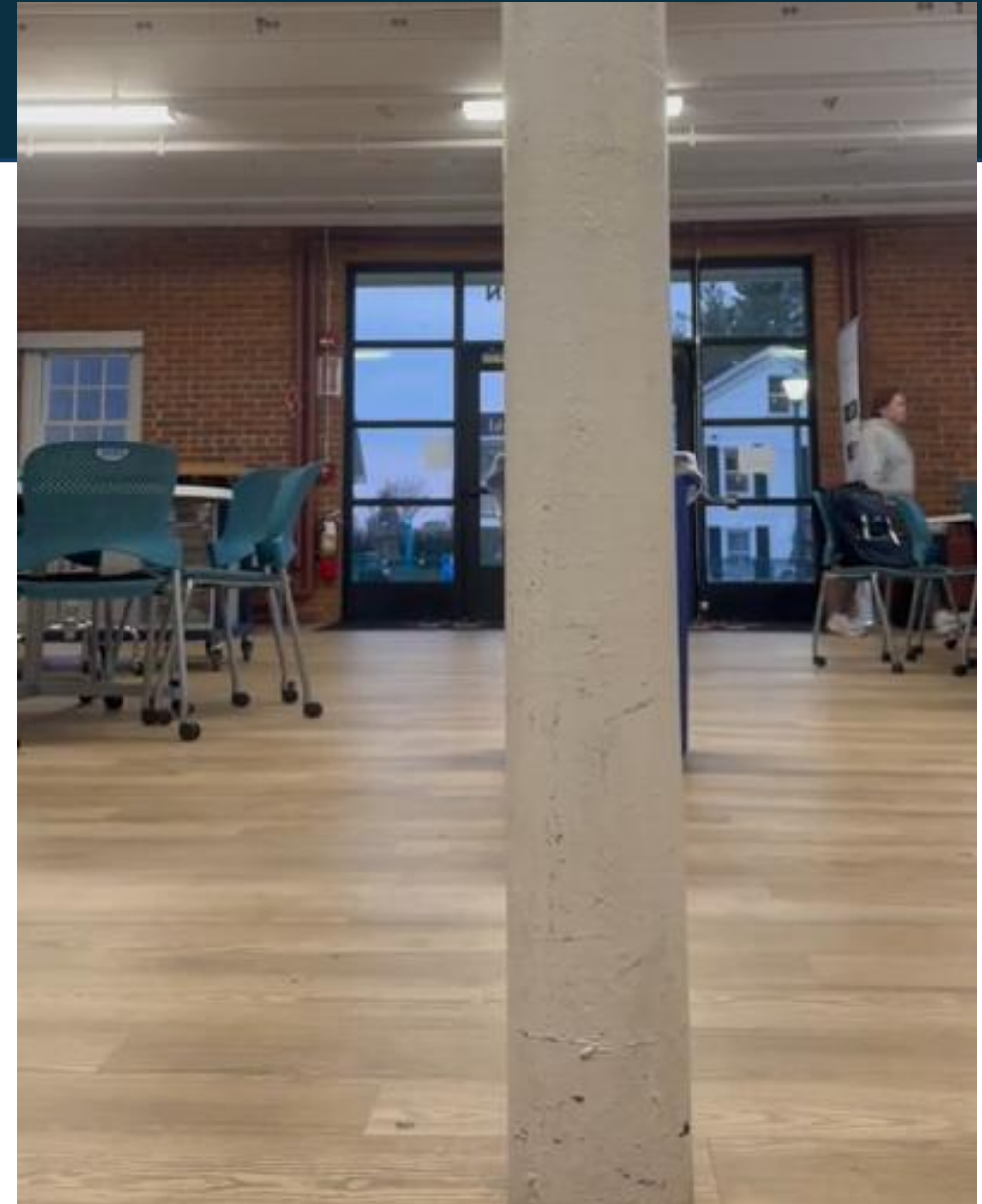
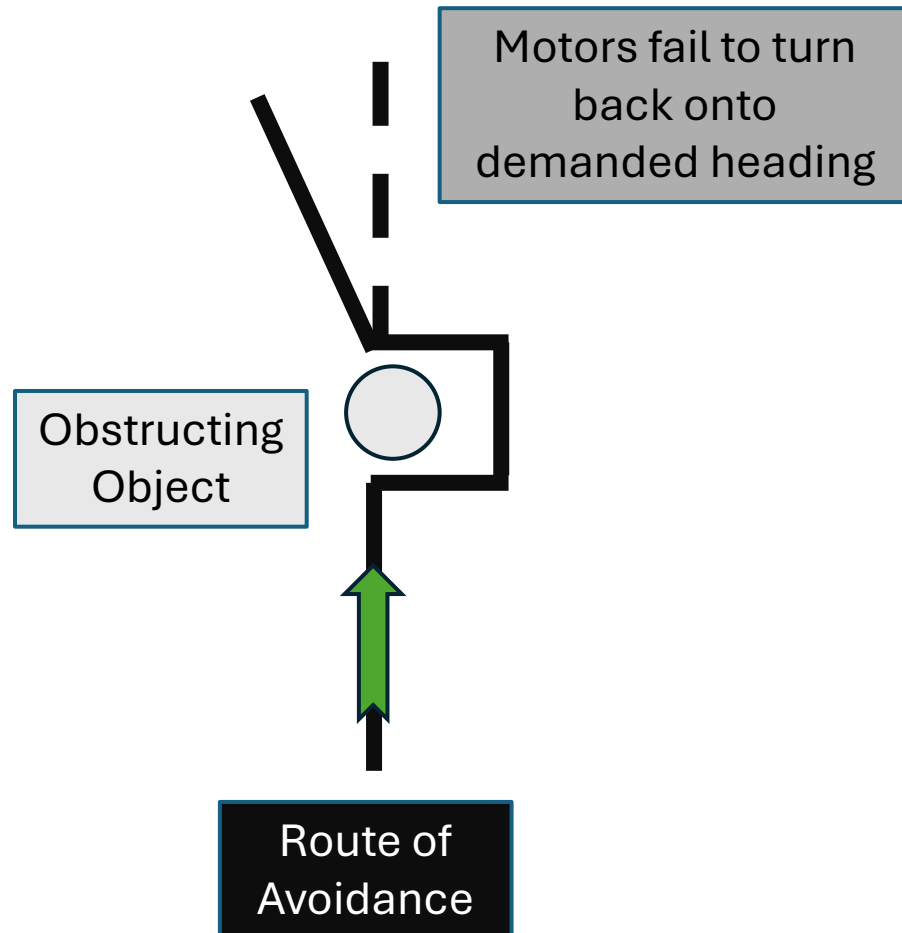
Initial Testing of Sensors



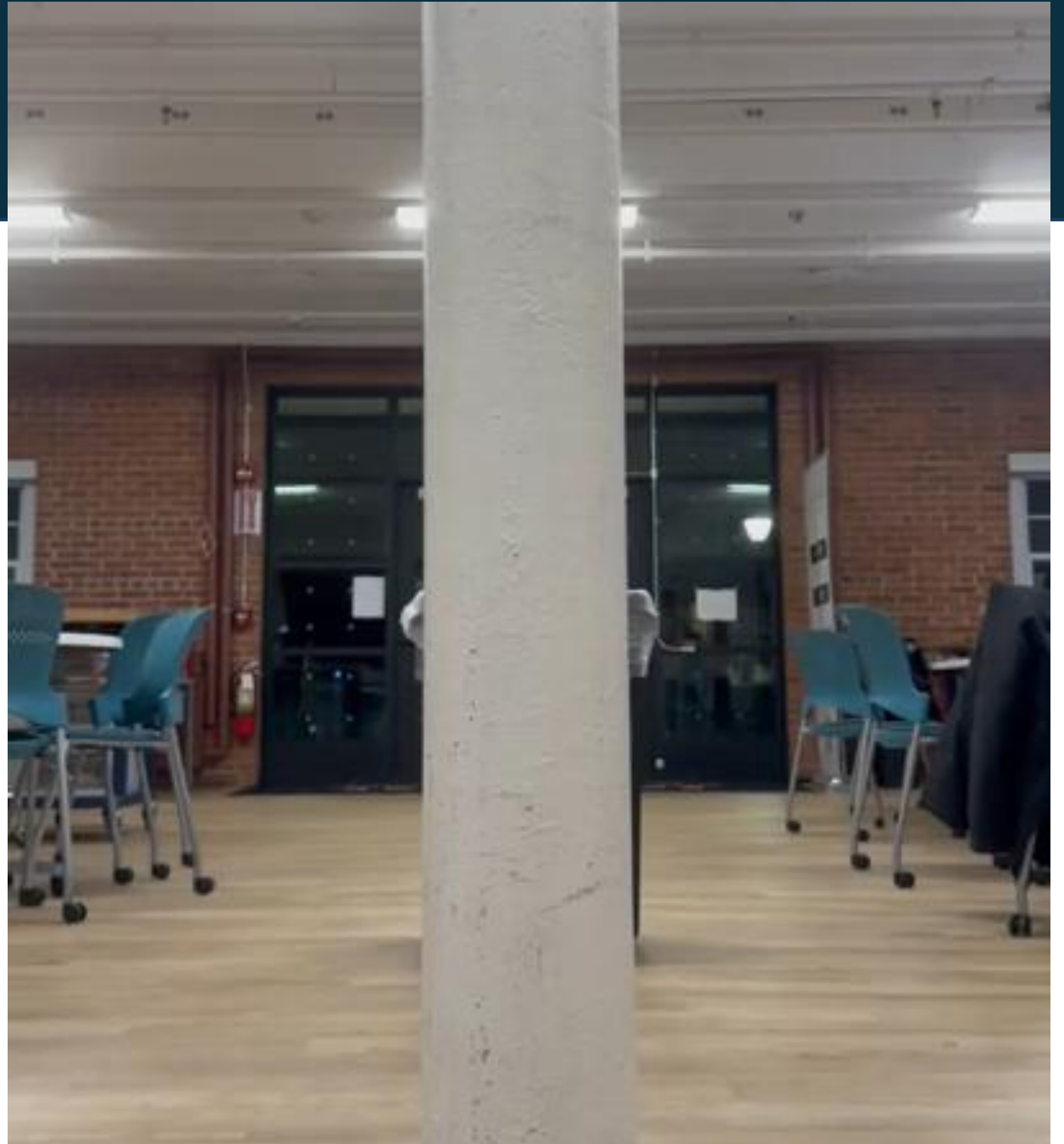
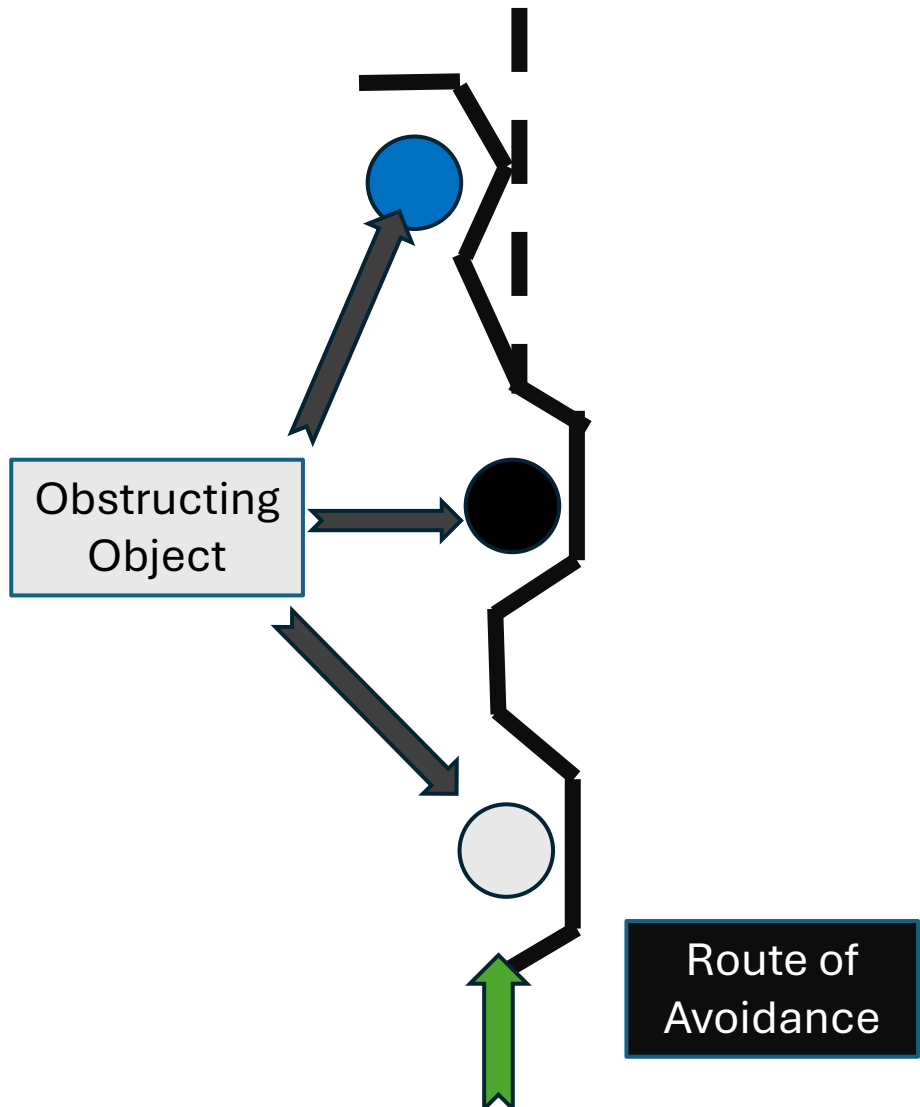
Logic Testing



Progress- POV of front sensor



POV of front sensor



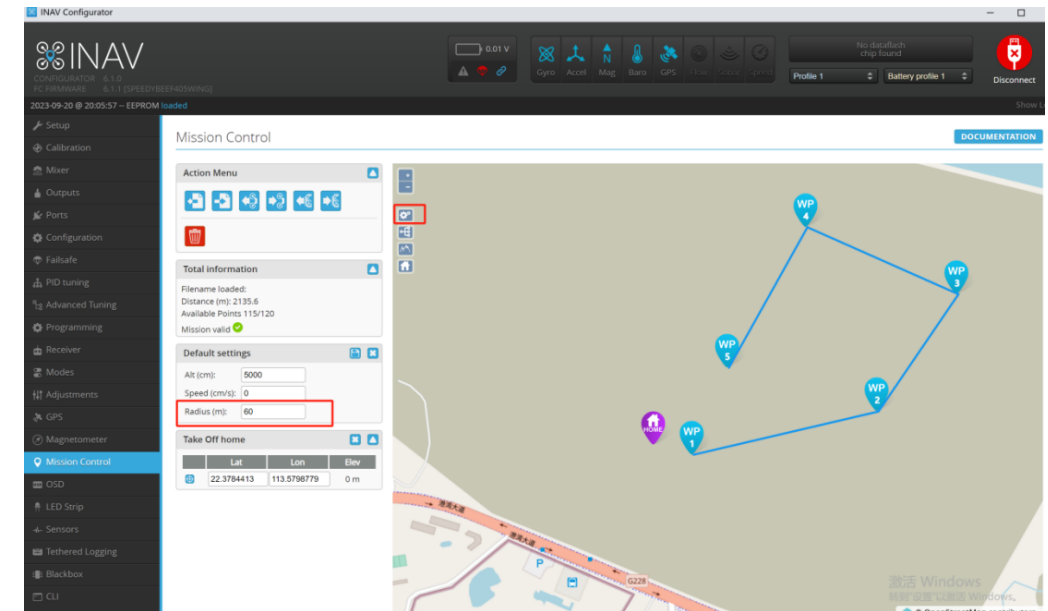
Risks/Mitigations

- **Brownout**
 - Reduced Power Draw
 - Better suited battery options
 - Voltage regulator
- **Inefficient Steering**
 - PWM Tuning
- **Path Divergence**
 - 90° Turns, executed the most reliably



Future Work

- **GPS integration**
 - Allows for coordinate style navigation, app interface to mark desired start/ stop positions, use photo data to outline driveway surface.
- **Higher torque motors**
 - Rated for heavy application, metal shaft motors
 - 12V-24V system (24V motor pictured)
- **Improved obstacle detection**
 - Logic works well for large obstructions, will not sense small, flat objects.



Thank You

Questions?